

✓ RAG with ChromaDB – Simple Google Colab Demo

This notebook demonstrates Retrieval Augmented Generation (RAG) using ChromaDB.

```
# Install necessary libraries: chromadb for the vector database and sentence-transformers for generating embeddings.
!pip install chromadb sentence-transformers
```

Collecting chromadb

```
Downloading chromadb-1.4.1-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.2 kB)
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.2.2)
Collecting build>=1.0.3 (from chromadb)
  Downloading build-1.4.0-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: pydantic>=1.9 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.12.3)
Collecting pybase64>=1.4.1 (from chromadb)
  Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64
Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.12/dist-packages (from uvicorn[standard]>=0.18.3-
Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.0.2)
Collecting posthog<6.0.0,>=2.4.0 (from chromadb)
  Downloading posthog-5.4.0-py3-none-any.whl.metadata (5.7 kB)
Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.15.0)
Collecting onnxruntime>=1.14.1 (from chromadb)
  Downloading onnxruntime-1.24.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (4.9 kB)
Requirement already satisfied: opentelemetry-api>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.37.0)
Collecting opentelemetry-exporter-otlp-proto-grpc>=1.2.0 (from chromadb)
  Downloading opentelemetry_exporter_otlp_proto_grpc-1.39.1-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.37.0)
Requirement already satisfied: tokenizers>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.22.2)
Collecting pypika>=0.48.9 (from chromadb)
  Downloading pypika-0.51.1-py2.py3-none-any.whl.metadata (51 kB)
52.0/52.0 kB 1.6 MB/s eta 0:00:00
Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.67.2)
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Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.76.0)
Collecting bcrypt>=4.0.1 (from chromadb)
  Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB)
Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.21.1)
Collecting kubernetes>=28.1.0 (from chromadb)
  Downloading kubernetes-35.0.0-py2.py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: tenacity>=8.2.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (9.1.2)
Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.0.3)
Requirement already satisfied: mmh3>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (5.2.0)
Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.12/dist-packages (from chromadb) (3.11.7)
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (13.9.4)
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Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-trans
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transform
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.9.
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Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.12/dist-packages (from build>=1.0.3->chromadb) (2
Collecting pyproject_hooks (from build>=1.0.3->chromadb)
  Downloading pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (2026.1.4
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Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20
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Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sente
```

```
documents = [
    "RAG stands for Retrieval Augmented Generation.",
    "ChromaDB is a vector database used to store embeddings.",
    "RAG reduces hallucinations by grounding answers using retrieved documents.",
    "Fine-tuning changes behavior, RAG injects knowledge at runtime.",
    "RAG retrieves relevant chunks before generation."
]
# Display the list of documents to be used.
documents
```

```
['RAG stands for Retrieval Augmented Generation.',
 'ChromaDB is a vector database used to store embeddings.',
 'RAG reduces hallucinations by grounding answers using retrieved documents.',
 'Fine-tuning changes behavior, RAG injects knowledge at runtime.',
 'RAG retrieves relevant chunks before generation.']
```

```
# Import the SentenceTransformer model for generating embeddings.
from sentence_transformers import SentenceTransformer
# Import the chromadb library for interacting with ChromaDB.
import chromadb

# Initialize the SentenceTransformer model with 'all-MiniLM-L6-v2' for creating embeddings.
model = SentenceTransformer('all-MiniLM-L6-v2')
# Generate embeddings for each document in the 'documents' list.
embeddings = model.encode(documents)

# Initialize the ChromaDB client.
client = chromadb.Client()
# Create a new collection named 'rag_demo' in ChromaDB.
collection = client.create_collection('rag_demo')

# Iterate through each document and its corresponding embedding.
for i, doc in enumerate(documents):
    # Add each document, its ID, and its embedding (converted to a list) to the ChromaDB collection.
    collection.add(ids=[str(i)], documents=[doc], embeddings=[embeddings[i].tolist()])

# Print a confirmation message once documents are stored.
print('Documents stored in ChromaDB')
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
modules.json: 100% 349/349 [00:00<00:00, 27.5kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 8.83kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
README.md: 10.5k/? [00:00<00:00, 821kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 5.07kB/s]

config.json: 100% 612/612 [00:00<00:00, 56.9kB/s]

model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 80.5MB/s]

Loading weights: 100% 103/103 [00:00<00:00, 477.84it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100% 350/350 [00:00<00:00, 16.1kB/s]

vocab.txt: 232k/? [00:00<00:00, 7.16MB/s]

tokenizer.json: 466k/? [00:00<00:00, 14.7MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 5.80kB/s]

config.json: 100% 190/190 [00:00<00:00, 9.92kB/s]
Documents stored in ChromaDB
```

```
# Define the query string.
query = 'How does RAG reduce hallucinations?'
# Generate the embedding for the query.
q_embed = model.encode([query])

# Perform a similarity search in the ChromaDB collection using the query embedding.
# Retrieve the top 2 most relevant results.
results = collection.query(query_embeddings=q_embed.tolist(), n_results=2)
# Extract and display the documents from the search results.
results['documents'][0]
```

```
['RAG reduces hallucinations by grounding answers using retrieved documents.',
 'RAG stands for Retrieval Augmented Generation.']
```

